THE 6D HOLOMORPHIC CHERN–SIMONS DEFECT AND THE POSITIVE HALF OF THE AFFINE YANGIAN

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Abstract. The local abelian $6d$ holomorphic Chern–Simons defect on $\mathbb{C}^3$ has a precise low-spin chiral content: a spin-one Heisenberg current with level $\Psi_{\mathrm{lev}} = -\sigma_2$, $\Psi_{\mathrm{lev}} \neq 0$, and its Sugawara spin-two field with central charge $c = 1$. It does not prove the all-spin $\mathcal{W}_{1+\infty}[\lambda]$ vertex algebra, and it does not identify the critical Hall theory of $\mathbb{C}^3$ with a vertex algebra. The spherical critical Hall algebra is the positive half $Y^+(\widehat{\mathfrak{gl}}_1)$; the $\mathcal{W}_{1+\infty}$ object belongs to the completed Drinfeld double and its generic Fock/mode realisation. Finite toric DWR/Ran Rees gluing assembles the chartwise positive halves. Compact critical CoHA, quasi-NCCR, and Hall–Drinfeld double statements require the extra realisation, compact-support, pairing, and completion data named below.

1. Defect OPE

The symbol Ψ_{lev} denotes the Heisenberg level of the defect current. It is not the Gaiotto–Rapčák corner coupling $\psi_{\mathrm{GR}} = -h_2/h_1$, which is dimensionless and controls the triality of the $Y_{L,M,N}[\psi_{\mathrm{GR}}]$ family. The parameter λ appearing in the Fock realisation below is a representation parameter for the completed double; no equality $\Psi_{\mathrm{lev}} = \psi_{\mathrm{GR}}$ or $\Psi_{\mathrm{lev}} = \lambda$ is used.

Theorem 1 (Low-spin abelian codimension-two witness)**.** *Let $C = \{z_2 = z_3 = 0\} \subset \mathbb{C}^3$ be a straight complex codimension-two defect in the local abelian $\mathfrak{gl}_1$ model of $6d$ holomorphic Chern–Simons theory. Let h_i be the Ω-background weights of the three coordinate directions, with $h_1 + h_2 + h_3 = 0$; only h_2, h_3 are normal weights to C. Work in a flat formal $\mathbb{C}^3$ chart, fix the framed defect and the current normalisation, choose the Ω-background regularisation and normal ordering, and use the five-dimensional-to-six-dimensional restriction convention over $\mathbb{C}[\Psi_{\mathrm{lev}}, \Psi_{\mathrm{lev}}^{-1}]$. Set*

$$\sigma_2 = h_1h_2 + h_1h_3 + h_2h_3, \qquad \Psi_{\mathrm{lev}} = -\sigma_2,$$

and assume $\Psi_{\mathrm{lev}} \neq 0$. The defect current $J(z)$ satisfies

$$J(z)J(w) \sim \frac{\Psi_{\mathrm{lev}}}{(z-w)^2}, \qquad \Psi_{\mathrm{lev}} = -\sigma_2.$$

The Sugawara field

$$T(z) = \frac{1}{2\Psi_{\mathrm{lev}}} :J(z)^2:$$

has central charge $c = 1$.

Proof. The BV fields of abelian hCS are the Dolbeault complex $\Omega^{0,*}(\mathbb{C}^3)[1]$ with the free differential $\bar\partial$ and the BV pairing induced by $dz_1 \wedge dz_2 \wedge dz_3$. The cubic hCS vertex is multiplied by the $\mathfrak{gl}_1$ bracket and vanishes. Hence the flat formal abelian model is Gaussian: its BRST cohomology on the straight defect contains the zero normal mode $A^{(0,0)}|_C$, and the only singular OPE of the corresponding current is the single Wick contraction of the BV propagator. No cubic Feynman graph is present, so the local calculation supplies no non-linear all-spin $\mathcal{W}_{1+\infty}$ relations.

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The free calculation determines a Heisenberg OPE once the trace form on the defect zero mode is fixed. The five-dimensional restriction normalisation, matched to the affine-Yangian classical r-matrix residue, fixes that trace form to be

$$k_{\mathrm{eff}} = -\sigma_2 = -(h_1h_2 + h_1h_3 + h_2h_3) = \frac{1}{2}(h_1^2 + h_2^2 + h_3^2),$$

where the last equality uses $h_1 + h_2 + h_3 = 0$. Thus the Heisenberg lambda-bracket is $\{J_\lambda J\} = \Psi_{\mathrm{lev}}\lambda$, equivalently the displayed double pole. Rescaling J would rescale the displayed parameter; the framed defect convention fixes J. At $\Psi_{\mathrm{lev}} = 0$ the singular part is zero and the Sugawara denominator is absent. For $\Psi_{\mathrm{lev}} \neq 0$, Wick contraction in the single Heisenberg algebra gives

$$T(z)T(w) \sim \frac{1/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w},$$

so $c = 1$. No higher-spin OPE is asserted. □

2. Hall algebra and completion

Theorem 2 (Toric critical Hall algebra)**.** *Let Q_3 be the one-vertex three-loop Jordan quiver with potential*

$$W = \mathrm{tr}(XYZ - XZY),$$

and let $\mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(Q_3, W)$ be the degree-one-generated spherical equivariant critical cohomological Hall algebra with vanishing cycles, the standard orientation choice, shifts, Tate twists, and equivariant parameters matched to (h_1, h_2, h_3). Then

$$\mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(Q_3, W) \cong Y^+(\widehat{\mathfrak{gl}}_1).$$

This is a spherical positive-half statement: it does not present the full non-spherical Hall algebra, the opposite half, the Cartan sector, or vertex-algebra locality. In the shuffle realisation, the n-particle summand is represented over $\mathbb{C}[h_1, h_2, h_3]/(h_1 + h_2 + h_3)$ by symmetric functions, and the product uses the positive Jordan kernel, with $u = x_j - x_i$,

$$\omega_+(u) = \frac{(u+h_1)(u+h_2)(u+h_3)}{u^3} = 1 + \frac{\sigma_2}{u^2} + \frac{\sigma_3}{u^3}, \qquad \sigma_3 = h_1h_2h_3.$$

The opposite extension convention replaces h_i by $-h_i$ and gives

$$\omega_-(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{u^3} = 1 + \frac{\sigma_2}{u^2} - \frac{\sigma_3}{u^3}.$$

After localisation, a Hopf pairing, quotient by its radical when nonzero, Cartan data, the opposite half, root-degree completion, and a generic non-resonant Fock parameter have been fixed, the completed Drinfeld double has a representation on the corresponding $\mathcal{W}_{1+\infty}[\lambda]$ vacuum module,

$$Y^+(\widehat{\mathfrak{gl}}_1) \hookrightarrow D^{\wedge}_{\mathrm{Hopf}}\big(Y^+(\widehat{\mathfrak{gl}}_1)\big) \xrightarrow{\mathrm{ev}_\lambda} \mathrm{End}(\mathcal{F}_\lambda).$$

The map ev_λ is the completed mode/enveloping action on $\mathcal{F}_\lambda$; it is not asserted to be onto $\mathrm{End}(\mathcal{F}_\lambda)$, nor to identify Y^+ or its double with the vertex algebra.

Proof. For a vector space V of dimension n, the critical stack is

$$[\mathrm{End}(V)^3/\mathrm{GL}(V)], \qquad W_n(X,Y,Z) = \mathrm{tr}(X[Y,Z]).$$

The vanishing-cycle sheaf φ_{W_n}, the orientation square root, and the conventional shifts and Tate twists are the data that make the extension-correspondence product homogeneous and associative. The Thom–Sebastiani isomorphism identifies the vanishing cycles on the extension correspondence with the external product of the two smaller vanishing-cycle sheaves. This gives an E_1 spherical critical CoHA; the ordered extension correspondence supplies no braiding.

Schiffmann–Vasserot and the Rapčák–Soibelman–Yang–Zhao comparison identify this spherical equivariant critical Hall algebra, in the above orientation and normalisation, with the shuffle presentation of the positive half $Y^+(\widehat{\mathfrak{gl}}_1)$. The displayed kernel is the first-principles remnant of the three loops and the three Jacobi relations of $W = \mathrm{tr}(X[Y,Z])$: numerator factors record the three loop weights, while the cubic denominator records the three zero-weight diagonal/Jacobi factors in the Jordan self-correspondence. For the positive convention,

$$(u+h_1)(u+h_2)(u+h_3) = u^3 + \sigma_2 u + \sigma_3.$$

Since $h_1 + h_2 + h_3 = 0$, the u^{-1} term vanishes and the expansion is $1 + \sigma_2 u^{-2} + \sigma_3 u^{-3}$. Reversing the extension order changes σ_3 to $-\sigma_3$, not σ_2; equivalently $\omega_-(u) = \omega_+(-u)$. The shifts and Tate twists alter grading and mixed-Hodge normalisation; they do not create the negative half, the Cartan current, or vertex-algebra locality.

The representation-theoretic passage therefore has extra hypotheses. A localised Hopf pairing gives the continuous dual negative half and the Cartan cross-relations; completing in the root grading gives the Drinfeld double. At generic parameters its Fock evaluation gives the completed mode action on the $\mathcal{W}_{1+\infty}[\lambda]$ vacuum representation. Thus $\mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(Q_3, W) = Y^+$, not $\mathcal{W}_{1+\infty}$. □

Definition 3 (Finite toric hCS–Hall comparison datum)**.** Let X_Σ be covered by finitely many toric CY_3 charts $U_\alpha \simeq \mathbb{C}^3/G_\alpha$. A finite local hCS–Hall comparison datum is a simplex-compatible cyclic critical atlas and a finite diagram of filtered E_1-algebras

$$\mathrm{Rees}_F\, \mathrm{Obs}^{\mathrm{Ran}}_{\mathrm{hCS}}(U_\alpha) \xrightarrow{\rho_\alpha} \mathrm{Rees}_F\, \mathcal{H}_\alpha, \qquad \mathcal{H}_\alpha := \mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(U_\alpha),$$

whose transition squares commute with the Hall wall-crossing quasi-isomorphisms and with the DWR/Ran face maps. Its assembled Hall output is the positive half

$$H^+_{X_\Sigma} := \mathrm{hocolim}_{\alpha\in\Sigma(3)}\, \mathcal{H}_\alpha.$$

Proposition 4 (Finite toric gluing stops at the positive half)**.** *The maps ρ_α assemble to a filtered E_1-map*

$$\rho^+_{X_\Sigma}\colon\ \mathrm{hocolim}_\alpha\, \mathrm{Rees}_F\, \mathrm{Obs}^{\mathrm{Ran}}_{\mathrm{hCS}}(U_\alpha) \longrightarrow \mathrm{Rees}_F\, H^+_{X_\Sigma}.$$

Thus the toric local hCS–Hall gluing exists at finite Rees level whenever the simplex-compatible cyclic atlas and orientation transports in the definition have been supplied. If the local completed doubles and pairing-preserving transition maps have also been constructed, they give a comparison morphism, where $D^\wedge$ denotes the completed Hopf double,

$$\mathfrak{o}_{\mathrm{dbl}}\colon\ \mathrm{hocolim}_\alpha\, D^\wedge(\mathcal{H}_\alpha) \longrightarrow D^\wedge(\mathrm{hocolim}_\alpha\, \mathcal{H}_\alpha),$$

whose failure to be an equivalence contains the failure of continuous dualisation

$$\mathrm{hocolim}_\alpha\, \mathcal{H}_\alpha^\vee \longrightarrow (\mathrm{hocolim}_\alpha\, \mathcal{H}_\alpha)^\vee$$

together with the failure of the local Hopf pairings to glue without radical. Hence finite local hCS–Hall comparison data do not construct a compact critical CoHA, a quasi-NCCR, a Hall–Drinfeld double, a PBW/no-extra-relations theorem, or $\mathcal{W}_{1+\infty}$.

Proof. The first map is the universal property of the finite homotopy colimit applied to the compatible filtered E_1-diagram. Drinfeld doubling is not a construction on a bare positive E_1-algebra: it uses a completed opposite half, continuous dualisation, Cartan data, and cross-relations from a Hopf pairing. Applying the double chartwise uses the chartwise continuous duals $\mathcal{H}_\alpha^\vee$; applying it after assembly uses $(\mathrm{hocolim}_\alpha\, \mathcal{H}_\alpha)^\vee$. These dual objects agree only under an exactness and pairing-gluing hypothesis. A compact CY_3 critical CoHA further requires oriented compact critical stacks, vanishing cycles with compact support, a coproduct compatible with compact support, a non-degenerate Hopf pairing after radical quotient, PBW/no-extra-relations compatibility for the completed double, and centre/Fock evaluation data. None of these is supplied by the finite spherical positive-half comparison map. □

3. Formal toroidal sector

Proposition 5 (Formal-disc toroidal algebra). *At the rank-one formal-disc shuffle level, with parameters*

$$q_i = \exp(h_i), \qquad q_1 q_2 q_3 = 1,$$

the Ding–Iohara–Miki presentation contains a compatible positive shuffle sector. With $z = \exp(\varepsilon u)$ *and* $q_i = \exp(\varepsilon h_i)$, *the positive multiplicative shuffle factor*

$$\prod_{i=1}^{3} \frac{1 - q_i z}{1 - z}$$

has rational limit $\omega_+(u)$ *as* $\varepsilon \to 0$. *The doubled DIM exchange function is the quotient of this positive factor by the opposite factor; it is not the positive Hall product itself. Miki's automorphism is an automorphism of the completed quantum toroidal algebra. The symmetry of the positive kernel gives external base-change isomorphisms under the cyclic relabelling*

$$(q_1, q_2, q_3) \longmapsto (q_2, q_3, q_1)$$

among positive shuffle algebras. This does not produce a fixed-parameter Miki automorphism, an $\mathfrak{S}_3$-*action, or triality on the positive affine-Yangian half. The internal Miki/triality symmetry of the* $\mathcal{W}_{1+\infty}$ *or corner-VOA side requires the doubled toroidal algebra, the opposite half, the Cartan sector, and the reflection involutions.*

Proposition 6 (Global surface lift obstruction). *For the local surface*

$$X = \mathrm{Tot}(K_{\mathbb{P}^1 \times \mathbb{P}^1})$$

or for the surface Hall algebra of $\mathbb{P}^1 \times \mathbb{P}^1$, *a compact Hall–Drinfeld or* $\mathcal{W}_{1+\infty}$ *statement requires a comparison*

$$\Theta_{\mathrm{hCS} \to \mathrm{Hall}}$$

from global hCS/Ran observables to the oriented critical Hall theory, together with the compact double data: orientation, compact support, compact-support-compatible coproduct, non-degenerate pairing after radical quotient, completion, negative half, Cartan, PBW/no-extra-relations compatibility, and centre/Fock hypotheses. Equivalently, the obstruction

$$\mathfrak{o}_{\mathrm{surf}} = (\mathfrak{o}_{\mathrm{orient}}, \mathfrak{o}_{\mathrm{supp}}, \mathfrak{o}_{\mathrm{coprod}}, \mathfrak{o}_{\mathrm{pair}}, \mathfrak{o}_{\mathrm{PBW}}, \mathfrak{o}_{\mathrm{rad}}, \mathfrak{o}_{\mathrm{Cartan}}, \mathfrak{o}_{\mathrm{cent}})$$

must vanish. A toric chartwise Ding–Iohara–Miki action supplies none of these global compact data by itself.

Remark 7 (Surface and local-threefold scopes). $\mathrm{Tot}(K_{\mathbb{P}^1 \times \mathbb{P}^1})$ is a non-compact toric CY_3; the surface Hall algebra of $\mathbb{P}^1 \times \mathbb{P}^1$ is a different two-dimensional Hall theory. Toric charts give local positive shuffle sectors in both pictures, but neither chart calculation supplies compact support, a global orientation datum, a non-degenerate Hopf pairing, or a PBW/no-extra-relations theorem for the completed double. The global lift obstruction is the simultaneous failure of these data to glue.